



## Answers/Solutions:

1. **B. 4 g**

### Solution:

1 Avogadro number means 1 mole of helium. The atomic mass of helium is 4 grams per mole. Therefore, 1 mole of helium weighs 4 g.

2. **D.  $5.485799 \times 10^{-4}$**

3. **A. neutrons**

4. **B. Electron affinity**

5. **C. Alkali metals**

6. **A. Group IIIA**

7. **B. Group VIA**

8. **D. Antimony**

9. **B. representatives**

10. **C. Fluorine**

11. **B. inversely proportional**

12. **D. Law of multiple proportion**

13. **A. Dalton's law of partial pressure**

14. **B. Graham's law of effusion**

15. **C. Henry's Law**

16. **B. 54.8 psia**

### Solution:

Note that a pressure gauge measures gauge pressure, that is, the pressure in excess of atmospheric pressure (1 atm = 14.7 psi). Thus, the initial pressure is 46.7 psi or 46.7 psia.

$$\frac{P_1}{P_2} = \frac{T_1}{T_2}$$

$$\frac{46.7}{P_2} = \frac{4+273}{50+273}$$

$$P_2 = 54.8 \text{ psia} \rightarrow \text{Ans}$$

17. **B. Molality**

18. **D. 10 L**

### Solution:

$N_2$ : 28 g/mol

$$V = 12.5 \text{ g } N_2 \times \frac{1 \text{ mol } N_2}{28 \text{ g } N_2} \times \frac{22.4 \text{ L } N_2}{1 \text{ mol } N_2}$$

$$V = 10 \text{ L } N_2 \text{ (STP)} \rightarrow \text{Ans}$$

19. **D. 240 g**

### Solution:

$$2.5 \text{ mol } Ca(NO_3)_2 \times \frac{6 \text{ mol O}}{1 \text{ mol } Ca(NO_3)_2}$$

$$\times \frac{16 \text{ g O}}{1 \text{ mol O}} = 240 \text{ g O} \rightarrow \text{Ans}$$

20. **C. 0.662 g**

### Solution:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$\frac{(740 \text{ torr})(230 \text{ ml})}{20^\circ\text{C} + 273} = \frac{(760 \text{ torr}) V_2}{0^\circ\text{C} + 273}$$

$$V_2 = 209 \text{ ml}$$

$Cl_2$ : 71 g/mol

$$m = 0.209 \text{ L} \times \frac{1 \text{ mol } Cl_2}{22.4 \text{ L}} \times \frac{71 \text{ g } Cl_2}{1 \text{ mol } Cl_2}$$

$$m = 0.662 \text{ g } Cl_2 \rightarrow \text{Ans}$$

21. **B. 6.6 g/L**

### Solution:

$$PV = nRT$$

$$PV = \frac{m}{MW} RT$$

$$\frac{m}{V} = \frac{P(MW)}{RT}$$

$$\frac{m}{V} = \frac{(3.5 \text{ atm})(44 \text{ g/mol})}{\left(0.0821 \frac{\text{Latm}}{\text{molK}}\right)(285 \text{ K})}$$

$$\frac{m}{V} = 6.6 \text{ g/L} \rightarrow \text{Ans}$$

22. **C. 0.00243**

### Solution:

$$PV = nRT$$

$$n = \frac{\left(\frac{755 \text{ mmHg}}{760 \text{ mmHg}}\right)(0.075 \text{ L})}{\left(0.0821 \frac{\text{Latm}}{\text{molK}}\right)(373 \text{ K})}$$

$$n = 0.00243 \text{ moles} \rightarrow \text{Ans}$$

23. **A. 128 g/mol**

### Solution:

$O_2$ : 32 g/mol

$$\frac{\text{Rate A}}{\text{Rate B}} = \frac{\sqrt{MW_B}}{\sqrt{MW_A}}$$

$$\frac{0.5V}{V} = \frac{\sqrt{32}}{\sqrt{MW}}$$

$$MW = 128 \text{ g/mol} \rightarrow \text{Ans}$$

24. **C. 1.2 atm**

### Solution:

Use Dalton's Law of Partial Pressure:

$$P = P_{He} + P_{Ar}$$

$$P = 0.45 \text{ atm} + 0.75 \text{ atm}$$

$$P = 1.20 \text{ atm} \rightarrow \text{Ans}$$

25. **B. 840 g**

### Solution:

$$P = 100 \text{ atm} \times \frac{1.013 \times 10^6 \frac{\text{dynes}}{\text{cm}^2}}{1 \text{ atm}}$$

$$P = 1.013 \times 10^8 \frac{\text{dynes}}{\text{cm}^2}$$

$$V = 50 \text{ L} \times \frac{1000 \text{ cm}^3}{1 \text{ L}}$$

$$V = 5 \times 10^4 \text{ cm}^3$$



$$PV = nRT$$

$$PV = \frac{m}{MW} RT$$

$$m = \frac{PV(MW)}{RT}$$

$$m = \frac{\left(1 \times 10^8 \frac{\text{dyne}}{\text{cm}^2}\right) \left(5 \times 10^4 \text{ cm}^3\right) \left(4 \text{ g/mol}\right)}{\left(8.3 \frac{\text{erg}}{\text{mol} \cdot \text{K}}\right) (288 \text{ K})}$$

$$m = 840 \text{ g} \rightarrow \text{Ans}$$

26. B. 0.058

**Solution:**

H<sub>3</sub>PO<sub>4</sub>: 96 g/mol

H<sub>2</sub>O: 18 g/mol

$$X_a = \frac{n_a}{n_a + n_b} \times 100\%$$

$$X_b = \frac{n_b}{n_a + n_b} \times 100\%$$

25% aqueous phosphoric acid solution means 25 g of H<sub>3</sub>PO<sub>4</sub> in 100 g of solution (25 g of H<sub>3</sub>PO<sub>4</sub> mixed with 75 g of water to form 100 g of solution)

$$X_{\text{H}_3\text{PO}_4} = \frac{n_{\text{H}_3\text{PO}_4}}{n_{\text{H}_3\text{PO}_4} + n_{\text{H}_2\text{O}}} \times 100\%$$

$$X_{\text{H}_3\text{PO}_4} = \frac{25 \text{ g} \left(\frac{1 \text{ mol}}{96 \text{ g}}\right)}{25 \text{ g} \left(\frac{1 \text{ mol}}{96 \text{ g}}\right) + 75 \text{ g} \left(\frac{1 \text{ mol}}{18 \text{ g}}\right)} \times 100\%$$

$$X_{\text{H}_3\text{PO}_4} = 5.9\%$$

27. B. 0.037 M

**Solution:**

C: 12 g/mol

H: 1 g/mol

O: 16 g/mol

MW of C<sub>6</sub>H<sub>12</sub>O<sub>6</sub> = 180 g/mol

$$M = \frac{\text{mol solute}}{\text{L solution}}$$

$$M = \frac{2.5 \text{ g} \times \frac{1 \text{ mol}}{180 \text{ g}}}{0.375 \text{ L}}$$

$$M = 0.037 \rightarrow \text{Ans}$$

28. C. 1.64 M

**Solution:**

MW of H<sub>3</sub>PO<sub>4</sub> = 98 g/mol

$$M = \frac{\text{mol solute}}{\text{L solution}}$$

$$M = \frac{102.9 \text{ g} \times \frac{1 \text{ mol}}{98 \text{ g}}}{0.642 \text{ L}}$$

$$M = 1.64 \rightarrow \text{Ans}$$

29. A. 7.59 g

**Solution:**

$$m = \frac{\text{mol solute}}{\text{kg solvent}}$$

$$0.35 = \frac{m \times \frac{1 \text{ mol}}{58.4 \text{ g}}}{0.750 \text{ kg}}$$

$$m = 7.59 \text{ g NaCl} \rightarrow \text{Ans}$$

30. D. 4.46 M

**Solution:**

Calcium chloride: CaCl<sub>2</sub>

MW = 111 g/mol

$$45\% = \frac{45 \text{ g CaCl}_2}{100 \text{ g soln}} \times 100\%$$

$$M = \frac{45 \text{ g CaCl}_2 \times \frac{1 \text{ mol}}{111 \text{ g}}}{100 \text{ g soln} \times \frac{1 \text{ L soln}}{1000 \text{ ml soln}}}$$

$$M = \frac{4.46 \text{ mol CaCl}_2}{1 \text{ L soln}}$$

31. D. 13

**Solution:**

$$\text{pH} = -\log[H^+]$$

$$\text{pH} = -\log[1 \times 10^{-3}]$$

$$\text{pH} = 13 \text{ (base)} \rightarrow \text{Ans}$$

32. C. SO<sub>3</sub>

**Solution:**

S: 32 g/mol

O: 16 g/mol

Assume that you have a 100 g sample, therefore you have 40 g of sulfur and 60 g of oxygen.

$$n_s = 40 \text{ g} \times \frac{1 \text{ mol}}{32 \text{ g}} = 1.25 \text{ mol}$$

$$n_o = 60 \text{ g} \times \frac{1 \text{ mol}}{16 \text{ g}} = 3.75 \text{ mol}$$

Divide each of the mole quantities by the smallest among them:

$$S: \frac{1.25 \text{ mol}}{1.25 \text{ mol}} = 1$$

$$O: \frac{3.75 \text{ mol}}{1.25 \text{ mol}} = 3$$

Therefore, the mole ratio is 1:3. Compound is SO<sub>3</sub> or Sulfur Trioxide.

33. B. Na<sub>2</sub>O<sub>2</sub>

**Solution:**

Molecular formula uses subscripts that report the actual number of atoms of each type in a molecule of the compound.

Get the molecular mass of NaO:

$$\text{MM} = 23 \frac{\text{g}}{\text{mol}} + 16 \frac{\text{g}}{\text{mol}}$$

$$\text{MM} = 39 \frac{\text{g}}{\text{mol}}$$

Divide the molecular mass of NaO to the actual mass given in the problem.



$$\frac{78 \frac{\text{g}}{\text{mol}}}{39 \frac{\text{g}}{\text{mol}}} = 2$$

The ratio is 2. Therefore, multiply each of the subscripts of NaO by 2 to get Na<sub>2</sub>O<sub>2</sub> or sodium peroxide.

34. **A. 48.43 %**

**Solution:**

$$\%O = \frac{(16)4}{132} \times 100\% = 48.4\% \rightarrow \text{Ans}$$

35. **B. C<sub>3</sub>H<sub>6</sub>O<sub>3</sub>**

**Solution:**

C: 40.0%, H: 6.71%, O: 53.3%

n = 0.320 mol

m = 28.8 g

$$MM_{\text{Total}} = \frac{m}{n} = \frac{28.8}{0.32} = 90 \frac{\text{g}}{\text{mol}}$$

$$MM_{\text{Total,C}} = 0.4(90) = 36 \frac{\text{g}}{\text{mol}}$$

$$\#C = \frac{MM_{\text{Total,C}}}{MM_C} = \frac{36}{12} \approx 3$$

$$MM_{\text{Total,H}} = 0.0671(90) = 6.039 \frac{\text{g}}{\text{mol}}$$

$$\#H = \frac{MM_{\text{Total,H}}}{MM_H} = \frac{6.039}{1} \approx 6$$

$$MM_{\text{Total,O}} = 0.533(90) = 47.97 \frac{\text{g}}{\text{mol}}$$

$$\#O = \frac{MM_{\text{Total,O}}}{MM_O} = \frac{47.97}{16} \approx 3$$

\ The compound is C<sub>3</sub>H<sub>6</sub>O<sub>3</sub>. → Ans

36. **D. 268.94°C**

**Solution:**

According to Charles' Law, volume is proportional to absolute temperature provided all other factors are held constant.

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}$$

$$\frac{5.5}{10} = \frac{25 + 273}{T_2 + 273}$$

$$T_2 = 268.94^\circ\text{C} \rightarrow \text{Ans}$$

37. **D. 8.65 x 10<sup>4</sup> g**

**Solution:**

$$\%U = \frac{71.4}{84.2} \times 100\% = 84.8\%$$

$$m_U = 0.848(102) = 86.5\text{kg} \rightarrow \text{Ans}$$

38. **B. P will increase by a factor of 4**

**Solution:**

According to Gay-Lussac's Law, pressure is proportional to the absolute temperature provided all other factors are held constant.

Ans. P will increase by a factor of 4

39. **D. P will increase by a factor of 2**

**Solution:**

According to Boyle's Law, pressure is inversely proportional to the volume provided all other factors are held constant.

Ans. P will increase by a factor of 2

40. **B. Feel slippery on the skin**

41. **C. 8**

42. **B. Effusion**

43. **C. Neutron**

44. **A. Intensive**

45. **C. The periodic law**

46. **B. It increases**

47. **B. Effusion**

48. **D. 2.5 %**

**Solution:**

$$\%H = \frac{1 \text{ g/mol}}{40 \text{ g/mol}} \times 100\% = 2.5\% \rightarrow \text{Ans}$$

49. **C. 16.0 kPa**

**Solution:**

$$\frac{n_O}{P_O} = \frac{n_O + n_N}{P_O + P_N}$$

$$\frac{0.5 \text{ mol}}{P_O} = \frac{0.5 \text{ mol} + 0.75 \text{ mol}}{40 \text{ kPa}}$$

$$P_O = 16 \text{ kPa} \rightarrow \text{Ans}$$

50. **A. 3.00**

**Solution:**

$$\frac{\text{rate Ar}}{\text{rate Cl}_2} = \sqrt{\frac{MW_{\text{Cl}_2}}{MW_{\text{Ar}}}} = \sqrt{\frac{70 \text{ g/mol}}{40 \text{ g/mol}}}$$

$$\frac{\text{rate Ar}}{\text{rate Cl}_2} = 1.32$$

$$\frac{4 \text{ mmol/minute}}{\text{rate Cl}_2} = 1.32$$

$$\text{rate Cl}_2 = 3 \text{ mmol/minute} \rightarrow \text{Ans}$$